

1.向量化计算

把数据转为宽表即可使用向量化计算，向量化计算相当于矩阵计算，省略了for loop大大加速了计算

窄表：行数较多列数较少，之前用过的所有数据都是以窄表形式储存

date	code	open	close	low	high	volume	money	turnover_ratio
2020/1/3	000001.XSH	2011.21	2039.71	2008.84	2055.14	940146	1.914E+09	0.5752
2020/1/3	000002.XSH	4511.83	4420.8	4383.56	4525.63	584000	2.584E+09	0.8292
2020/1/3	000004.XSH	166.12	166.42	164.04	167.39	179695	29779318	1.6198
2020/1/3	000005.XSH	32.1	32.2	31.9	32.2	800063	25660053	0.7731
2020/1/3	000006.XSH	248.58	252.27	248.12	255.03	359846	90465551	1.2286
2020/1/3	000007.XSH	115.25	114.64	114.52	115.49	278378	31958869	1.0862
2020/1/3	000008.XSH	99.67	100.47	99.13	100.47	1159790	115673791	1.198
2020/1/3	000009.XSH	66.66	64.06	63.86	67.46	17711443	1.155E+09	6.9689
2020/1/3	000010.XSH	45.33	45.33	45.06	45.46	149063	6739342	0.3795
2020/1/3	000011.XSH	35.73	35.65	35.35	35.88	804221	28593489	0.5739
2020/1/3	000012.XSH	161.48	158.94	157.67	161.48	609104	96790302	0.9891
2020/1/3	000014.XSH	85.96	86.05	85.41	86.33	155263	13330049	0.7032
2020/1/3	000016.XSH	105.51	105.27	104.56	106.69	1159868	122290317	1.7147
2020/1/3	000017.XSH	14.94	14.97	14.81	15.07	1620344	24203948	1.7402
2020/1/3	000018.XSH	1.96	1.96	1.89	2.03	7622429	14993745	5.5498
2020/1/3	000019.XSH	33.61	34.48	33.56	34.86	2306140	78891674	3.011
2020/1/3	000020.XSH	26.31	26.41	26.05	27.18	761993	20267599	1.0107
2020/1/3	000021.XSH	205.43	223.91	202.42	223.91	8806078	1.87E+09	9.4705
2020/1/3	000023.XSH	35.2	35.76	35.12	36.08	1343949	47862820	2.5924
2020/1/3	000025.XSH	58.01	57.79	57.79	58.5	1758863	102209872	1.2235
2020/1/3	000026.XSH	96.91	95.97	94.85	98.29	2747002	265656000	6.6045
2020/1/3	000027.XSH	112.77	113.14	112.05	113.14	437749	49311501	0.1996
2020/1/3	000028.XSH	179.84	181.77	179.52	184.44	507942	92618351	0.6495
2020/1/3	000030.XSH	15.94	16.01	15.84	16.08	918846	14669150	0.1734
2020/1/3	000031.XSH	136.03	134.72	132.87	136.03	1244673	166871795	1.2752
2020/1/3	000032.XSH	61.1	61.47	60.31	61.85	989244	60482803	0.9984
2020/1/3	000034.XSH	62.39	62.27	61.45	64.21	7193254	450649246	4.4789

宽表：在截面选股策略中指index为时间，columns为股票标的的数据

	trade_time	000001.SZ	000002.SZ	000004.SZ	000005.SZ	000006.SZ	000007.SZ	000008.SZ	000009.SZ	000010.SZ	...	688787.SH	688788.SH	688789.SH	688793.SH
13149	2021-01-04 09:35:00	18.818752	28.106561	20.604137	2.524095	5.593543	6.270000	2.526688	7.528392	4.164988	...	NaN	135.174076	NaN	NaN
7942	2021-01-04 09:40:00	18.568108	27.920482	20.667872	2.515743	5.559550	6.270000	2.516586	7.557622	4.049356	...	NaN	134.382772	NaN	NaN
22627	2021-01-04 09:45:00	18.556963	27.911063	20.813928	2.523432	5.561666	6.270000	2.517260	7.587267	4.060129	...	NaN	136.333662	NaN	NaN
7348	2021-01-04 09:50:00	18.698916	27.888455	20.812399	2.520985	5.574471	6.270000	2.521047	7.581827	4.046134	...	NaN	137.454506	NaN	NaN
9147	2021-01-04 09:55:00	18.622740	27.834236	20.910413	2.511270	5.562428	6.270000	2.515780	7.558496	4.051465	...	NaN	138.518409	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
125	2022-12-30 14:40:00	13.166788	18.228533	9.620000	1.870000	6.327982	7.762162	2.301445	12.043069	3.759258	...	59.797000	36.213298	171.933382	49.687161

窄表和宽表分别对应了两种完全不同的数据储存方案（按日期/股票存储vs按照字段存储）

窄表转宽表操作：concat/pivot

2.多进程原理讲解

多线程vs多进程

多线程特点：多个线程共享同一内存，python GUI全局锁导致python基本不用多线程

多进程特点：不共享内存，每个进程创建一份新的数据，所以占用内存较大

回测改良：on bar generate + 多进程 + 向量化运算